

Homework 27: Matrix Algebra

BCS A, E, F, G

Problem.

a. Understanding Matrix Multiplication. Which row(s) and column(s) of matrices A and B do you multiply to find the following:

- i. The third column of AB .
- ii. The third row of AB .
- iii. The entry in row 3, column 4 of AB .

b. Regarding Invertibility. Is it necessary for the sum of two invertible matrices to be invertible? Explain.

c. Cramer's Rule.

Consider the following system in which s is an unspecified parameter. Determine the values of s for which the system has a unique solution and use Cramer's rule to determine the solution.

$$3sx_1 - 2x_2 = 4, \quad -6x_1 + sx_2 = 1.$$

d. Linear Transformations and Determinants. Let S be the parallelogram determined by the vectors $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{b}_2 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, and let $A = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}$. Compute the area of the image of S under the mapping $x \rightarrow Ax$.

e. Inverse Through Elementary Matrices. For elementary matrices E_1 , E_2 , and E_3 , one has $E_1E_2E_3A = I$, where

$$E_1 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}.$$

Find the matrix A^{-1} .

Submission Date: October 26, 2024 (Saturday – 11:59PM)

Good Luck